

**Practical File
Of
PC Assembly & Troubleshooting Laboratory
(UGCA1920)**



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Computer Network Practical File Index Format

Sr. No.	Name of the Experiment/Program	Page No.	Teacher's Signature
1	Assembling and De Assembling of Computer System	3	
2	Loading and configuration procedure of Microsoft Client O/S Win XP /Win 7 and Windows 8	6	
3	Installation of utility tools (Software)	9	
4	Installation of utility tools (Drivers)	11	
5	Firewall configuration, Antivirus/Internet security loading and configuration procedure	13	
6	Installation and configuration of I/O devices – Printers, Webcams, Scanners	17	
7	Installation and configuration of I/O devices – Digital Camera, USB Wi-Fi, USB BT, USB Storages, Projectors	20	
8	Multiple OS loading and trouble shooting	25	

1. Assembling and De Assembling of Computer System.

Aim

To study the main components of a computer system and to learn the procedure of assembling and de-assembling a computer.



Requirements / Tools

- Computer cabinet (CPU cabinet)
- Motherboard
- Processor (CPU)
- Heat sink and cooling fan
- RAM
- Hard Disk / SSD
- SMPS (Power Supply)
- Connecting cables
- Screwdriver

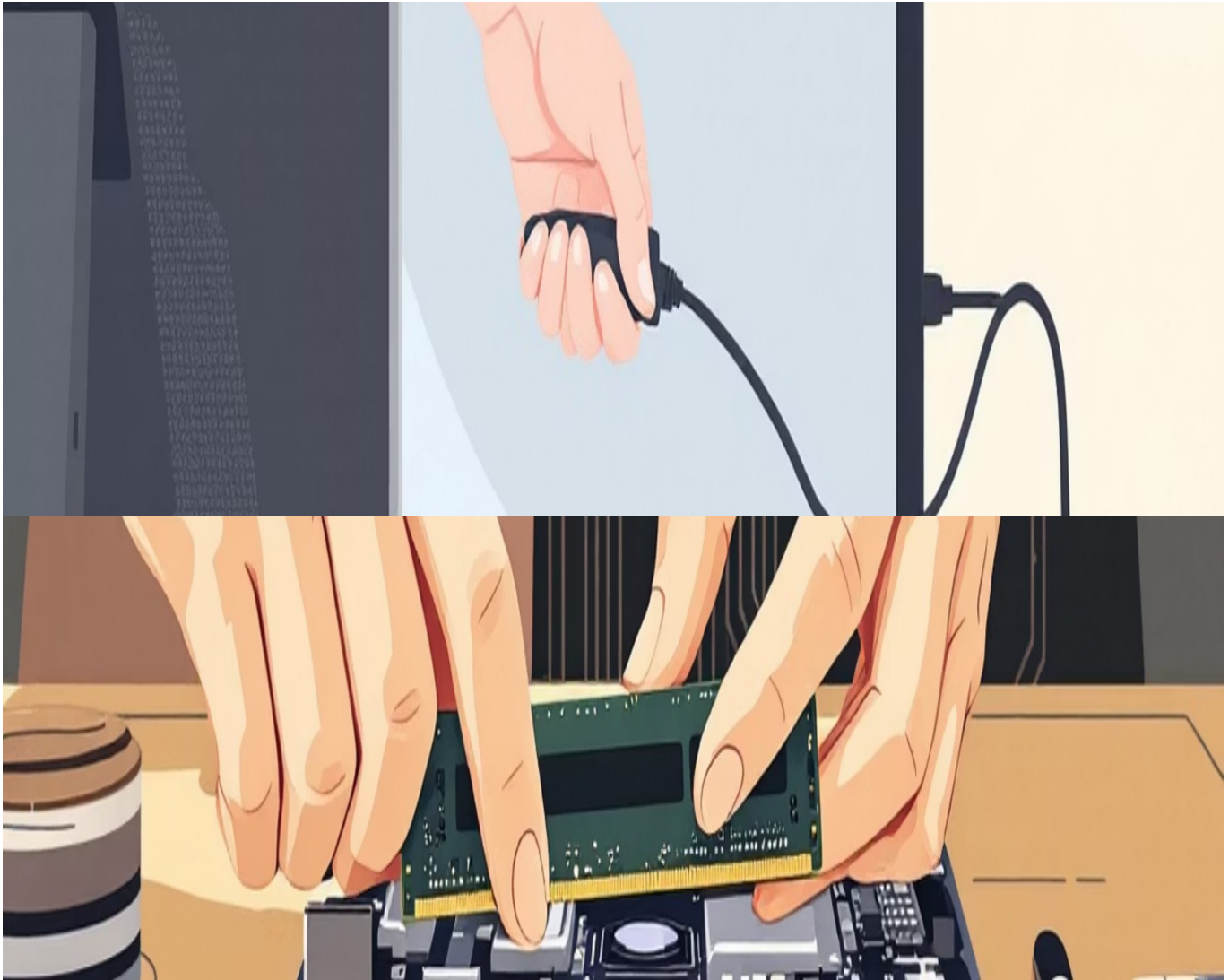
Theory

Assembling a computer means fitting all internal hardware components together to make a working computer system. De-assembling means safely removing these components for repair, upgrade, or study purposes.

Procedure for De-Assembling

1. Switch off the computer and unplug all power cables.

2. Open the CPU cabinet using a screwdriver.
3. Disconnect power and data cables from hard disk and motherboard.
4. Remove RAM carefully by opening the side clips.
5. Remove the hard disk/SSD from its slot.
6. Remove the cooling fan and processor carefully.
7. Unscrew and remove the motherboard from the cabinet.



Procedure for Assembling

1. Fix the SMPS in the cabinet.
2. Place the motherboard in the cabinet and screw it properly.
3. Insert the processor into the CPU socket.
4. Attach the heat sink and cooling fan.
5. Insert RAM into the memory slot.
6. Fix hard disk/SSD in the drive bay.
7. Connect power and data cables to all components.
8. Close the cabinet and connect external devices.

9. Switch on the system and check proper working.



Result

The computer system was successfully de-assembled and assembled, and all components were identified and connected correctly.

Precautions

- Always switch off and unplug the system before opening the cabinet.
- Handle components carefully to avoid damage.
- Do not touch electronic parts with wet hands.

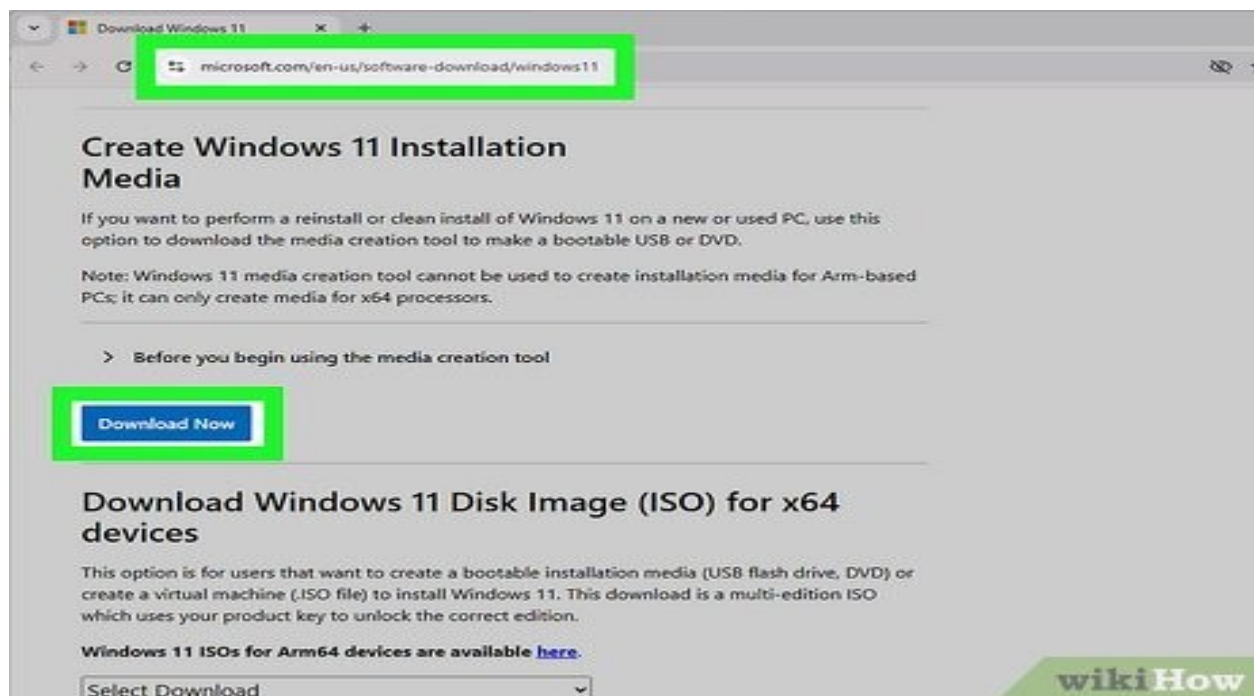
2. Loading and configuration procedure of Microsoft Client O/S Win XP /Win 7 and Windows 8.

Aim

To study and perform the loading (installation) and basic configuration of Microsoft Client Operating System such as Windows XP, Windows 7, and Windows 8.

Requirements

- Computer system
- Windows XP / Windows 10 / Windows 11 installation DVD or bootable USB
- Keyboard, mouse, monitor
- Hard disk



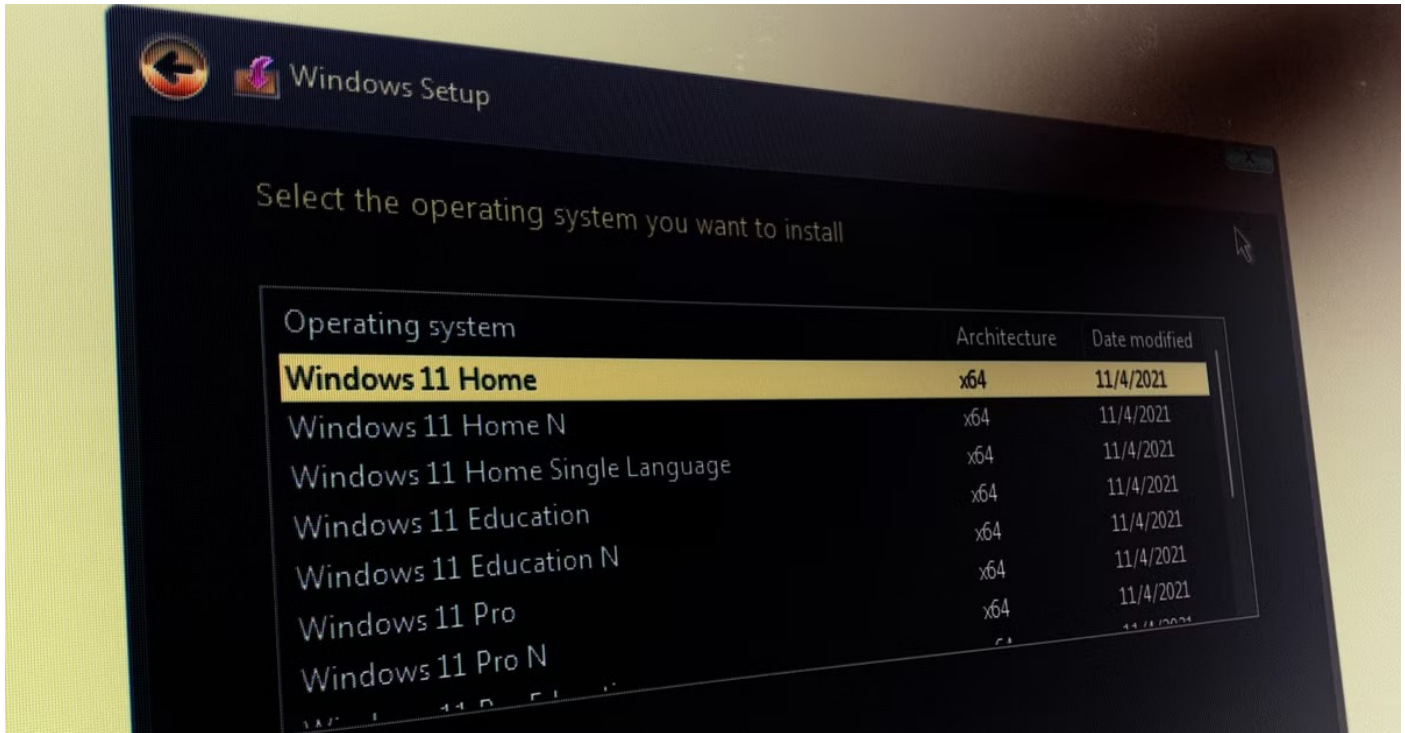
Theory

Loading an Operating System means installing the OS on a computer so that it can manage hardware and software resources. Configuration means setting up system settings such as user account, date and time, display, and device drivers after installation.

Procedure for Loading (Installation) of Windows OS

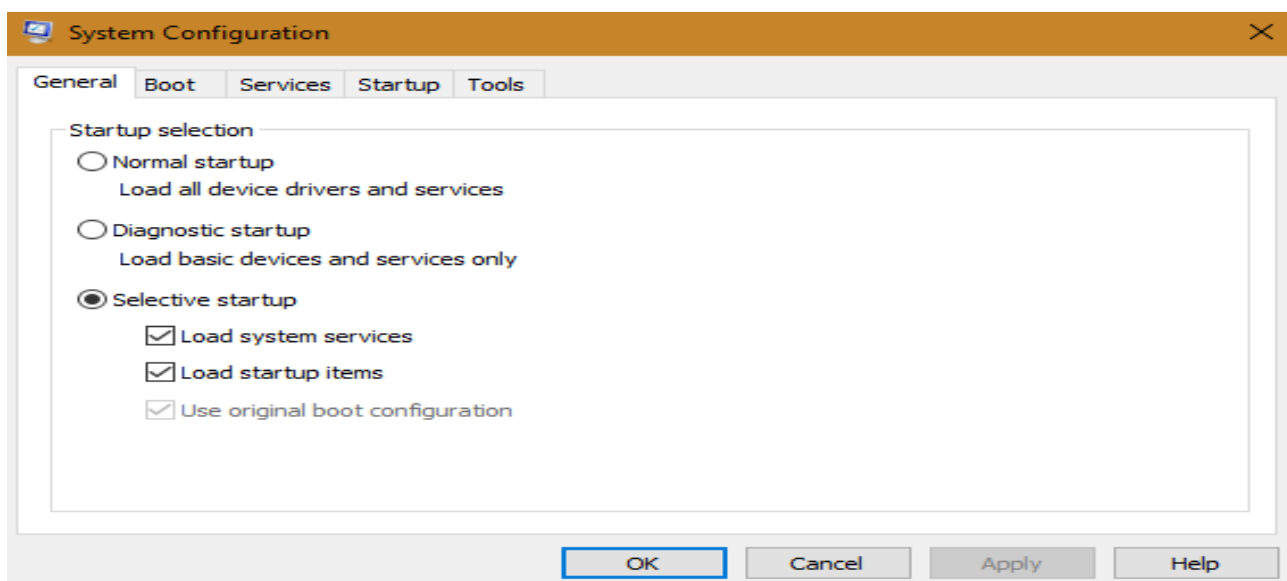
1. Insert Windows installation DVD or connect bootable USB.
2. Restart the computer and enter BIOS/UEFI setup.
3. Set boot priority to DVD/USB.
4. Save settings and restart the system.
5. Press any key to boot from DVD/USB.
6. Select language, time, and keyboard settings.
7. Click **Install Now**.
8. Select partition and format if required.

9. Windows files are copied and system restarts automatically.
10. Installation process completes successfully.



Procedure for Configuration of Windows OS

1. Enter user name and computer name.
2. Set date and time.
3. Select display resolution.
4. Install required device drivers.
5. Configure network and internet settings.
6. Set password and security options.
7. Activate Windows (if required).



Result

Microsoft Client Operating System (Windows XP / Windows 7 / Windows 8) was successfully loaded and configured on the computer system.

Precautions

- Backup important data before installation.
- Do not turn off the system during installation.
- Use original or genuine Windows OS.

3. Installation of utility tools (Software)

Aim

To study and perform the installation of utility software tools on a computer system.

Requirements

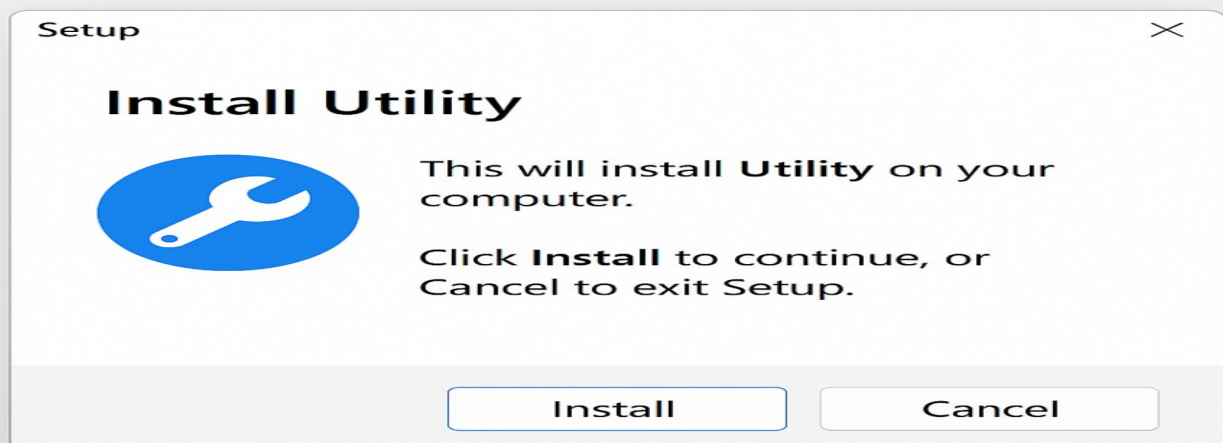
- Computer system with Windows OS
- Utility software (Antivirus, Disk Cleanup, Compression tool, Media Player, etc.)
- Installation CD/DVD or setup file
- Mouse and keyboard

Theory

Utility tools are system software that help in maintaining, managing, and protecting a computer system. These tools improve system performance, security, and efficiency. Examples include antivirus software, disk cleanup tools, file compression tools, and backup utilities.

Procedure for Installation of Utility Software

1. Switch on the computer system.
2. Copy or insert the utility software setup file (CD/DVD/USB).
3. Double-click on the setup (.exe) file.
4. Click **Next** on the installation wizard.
5. Accept the license agreement.
6. Choose installation location (default or custom).
7. Click **Install** to start installation.
8. Wait until installation is completed.
9. Click **Finish** to complete the process.



Examples of Utility Tools

- Antivirus Software (Quick Heal, Avast, Windows Defender)
- Disk Cleanup Tool
- Disk Defragmenter
- File Compression Tools (WinZip, WinRAR)
- Backup Utility

Result

The utility tool software was successfully installed and is working properly on the computer system.

Precautions

- Install software from a trusted source.
- Do not interrupt the installation process.
- Check system requirements before installation.

4. Installation of utility tools (Drivers)

Aim

To study and perform the installation of device drivers required for proper functioning of computer hardware.

Requirements

- Computer system with Windows OS
- Device driver software (CD/DVD or downloaded drivers)
- Keyboard, mouse, monitor
- Internet connection (optional)

Theory

Device drivers are special utility programs that allow the operating system to communicate with hardware devices such as printer, sound card, graphics card, network card, and motherboard. Without drivers, hardware devices cannot work properly.

Procedure for Installation of Drivers

1. Switch on the computer system.
2. Insert the driver CD/DVD or open the downloaded driver folder.
3. Double-click on the driver setup file.
4. Click **Next** on the installation wizard.
5. Accept the license agreement.
6. Click **Install** to begin installation.
7. Wait until the driver installation is completed.
8. Restart the computer if required.



Examples of Device Drivers

- Printer Driver
- Sound Driver
- Graphics (Display) Driver
- Network (LAN/Wi-Fi) Driver
- USB Driver

Result

The required device drivers were successfully installed and the hardware devices are working properly.

Precautions

- Install correct driver for the operating system.
- Do not interrupt installation process.
- Restart the system after driver installation.

5. Firewall configuration, Antivirus/Internet security loading and configuration procedure

Aim:

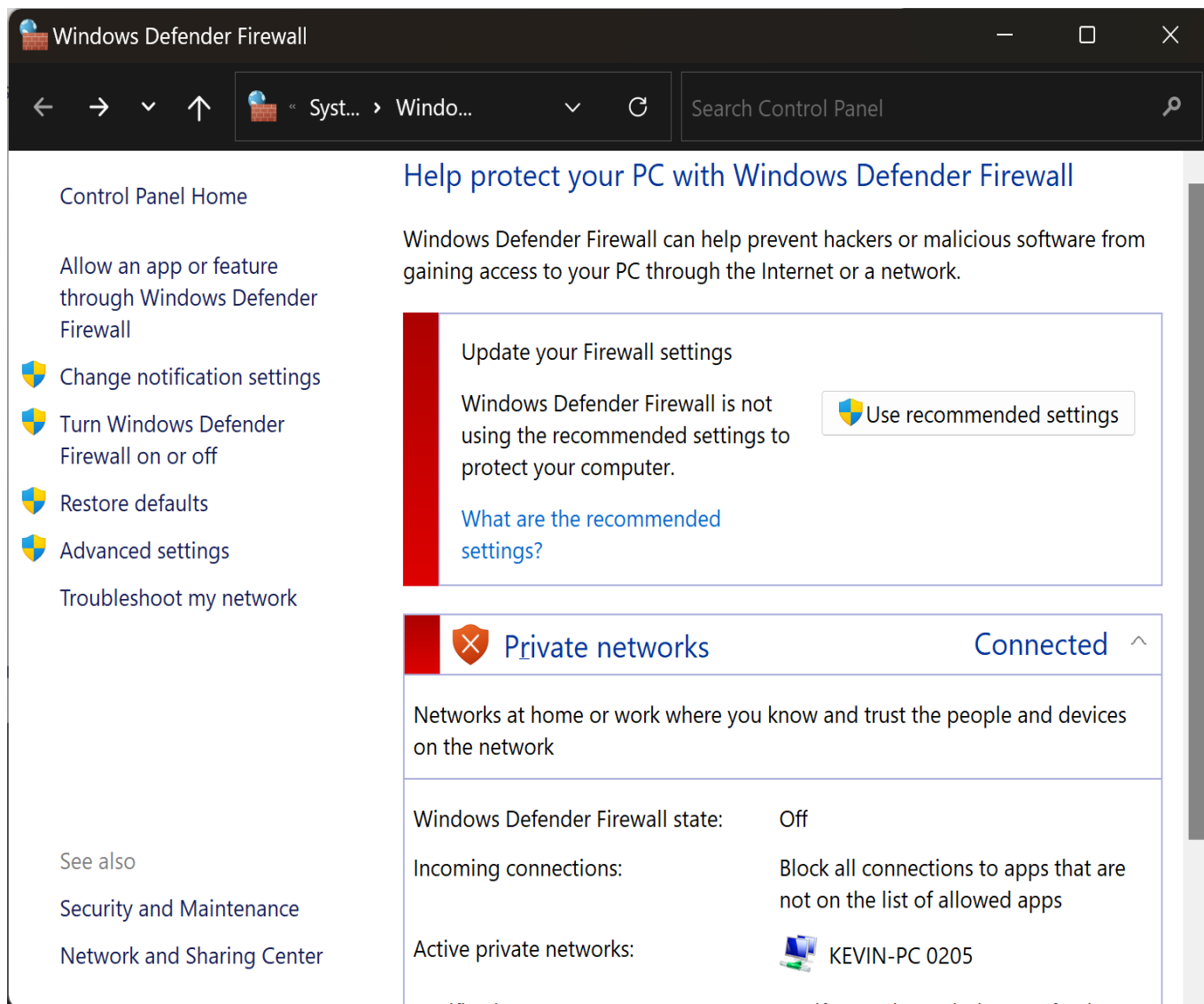
To configure firewall settings to protect the computer from unauthorized access.

Requirement:

Computer system, Windows OS, Internet connection.

Procedure:

1. Click on **Start Menu** → **Control Panel**.
2. Open **Windows Defender Firewall**.
3. Click **Turn Windows Defender Firewall on or off**.
4. Select **Turn on Firewall** for:
 - Private Network
 - Public Network
5. Click **OK**.
6. To allow an app:
 - Click **Allow an app through firewall**.
 - Select required application and click **OK**.



Result:

Firewall is successfully configured and the system is protected.

B. Antivirus / Internet Security Loading and Configuration

Aim:

To install and configure antivirus software for system security.

Requirement:

Antivirus software (Quick Heal / Avast / Windows Defender), Computer system.

Procedure:

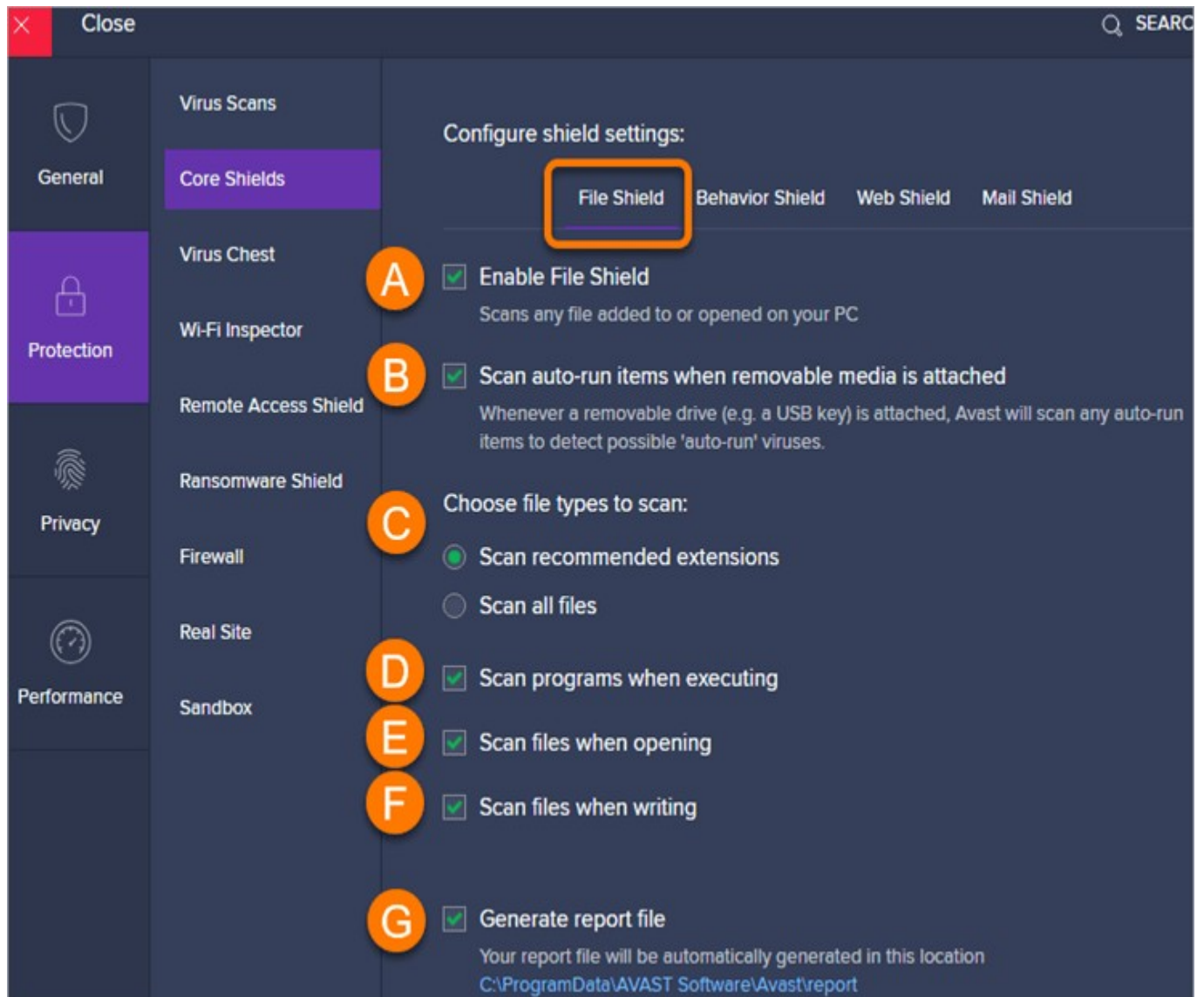
1. Insert antivirus CD / Download antivirus setup file.
2. Double-click on **Setup.exe**.
3. Click **Install** and accept terms & conditions.
4. Wait for installation to complete.
5. Restart the computer if required.
6. Open antivirus software.

7. Update virus definitions.

8. Enable:

- Real-time protection
- Auto scan
- Internet security

9. Run **Full System Scan**.



Result:

Antivirus software is successfully installed and configured.



Conclusion:

Firewall and antivirus protect the system from viruses, hackers, and unauthorized access.

6. Installation and configuration of I/O devices – Printers, Webcams, Scanners.

Aim:

To install and configure input/output devices such as Printer, Webcam, and Scanner.

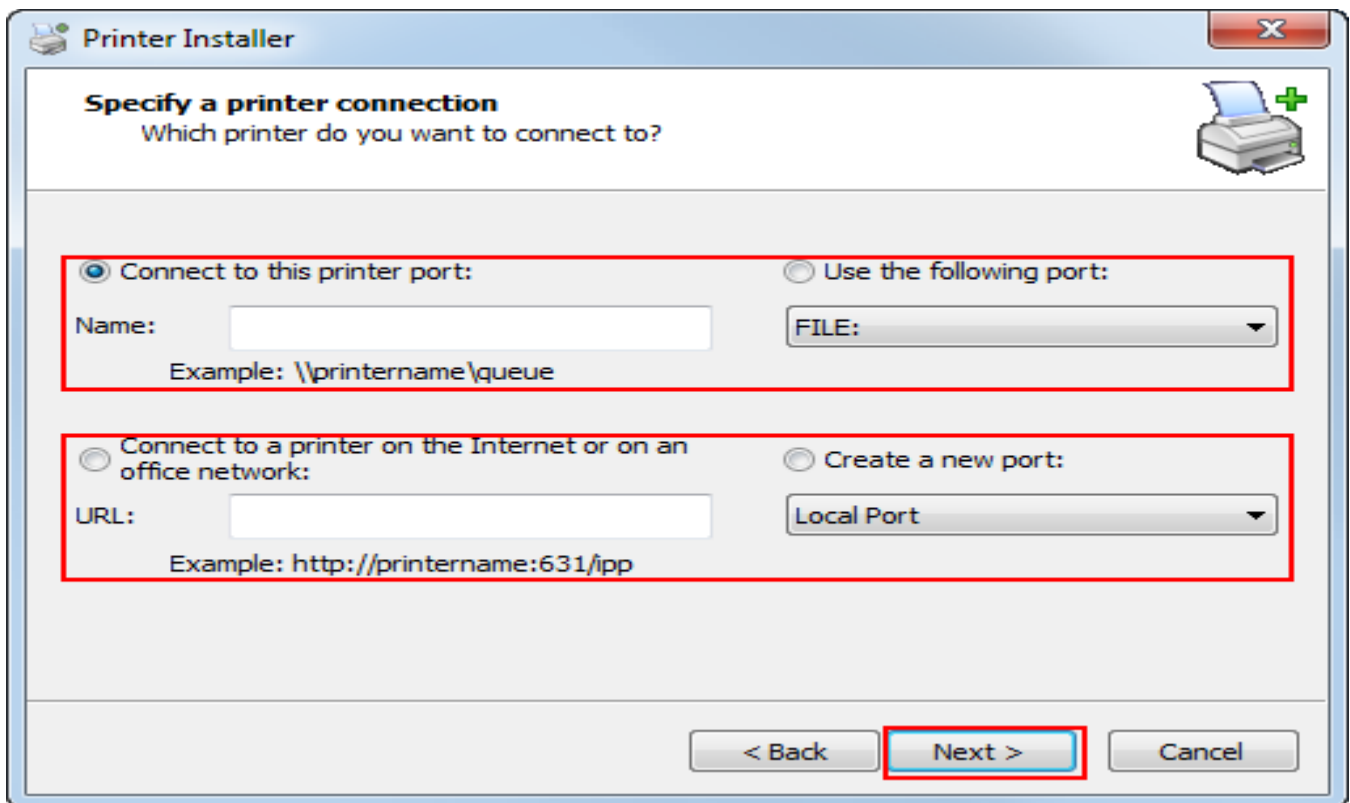
Requirement:

Computer system, Printer, Webcam, Scanner, Device drivers, USB cables.

A. Installation and Configuration of Printer

Procedure:

1. Connect the printer to the computer using USB cable.
2. Switch on the printer.
3. Install printer driver from CD or download from website.
4. Go to **Control Panel** → **Devices and Printers**.
5. Click **Add a Printer**.
6. Select the printer and click **Next**.
7. Set printer as **Default Printer**.
8. Print a **Test Page**.



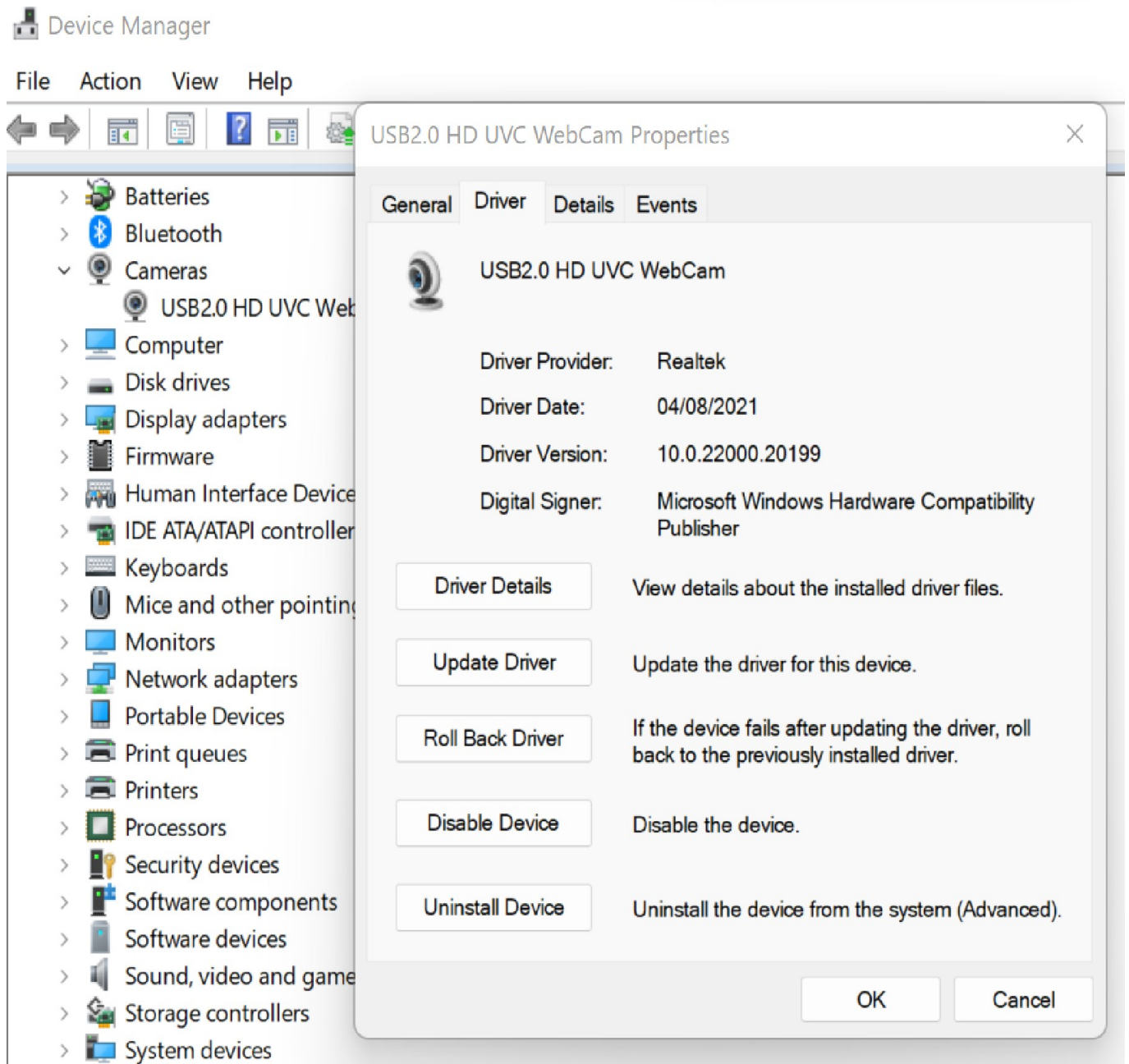
Result:

Printer is successfully installed and configured.

B. Installation and Configuration of Webcam

Procedure:

1. Connect the webcam to the USB port.
2. Windows automatically installs the driver.
3. If not installed, install driver manually.
4. Open **Camera App** or video software.
5. Adjust brightness, focus, and resolution.
6. Check video output.



Result:

Webcam is installed and working properly.

C. Installation and Configuration of Scanner

Procedure:

1. Connect scanner to the computer using USB cable.

2. Turn on the scanner.
3. Install scanner driver/software.
4. Open scanning software.
5. Place document on scanner glass.
6. Select scan type and resolution.
7. Click **Scan** and save the file.



Result:

Scanner is successfully installed and configured.

Conclusion:

Printers, webcams, and scanners are important I/O devices. Proper installation and configuration ensure smooth operation.

7. Installation and configuration of I/O devices – Digital Camera, USB Wi-fi, USB BT, USB Storages, Projectors

Aim:

To install and configure various I/O devices used with a computer system.

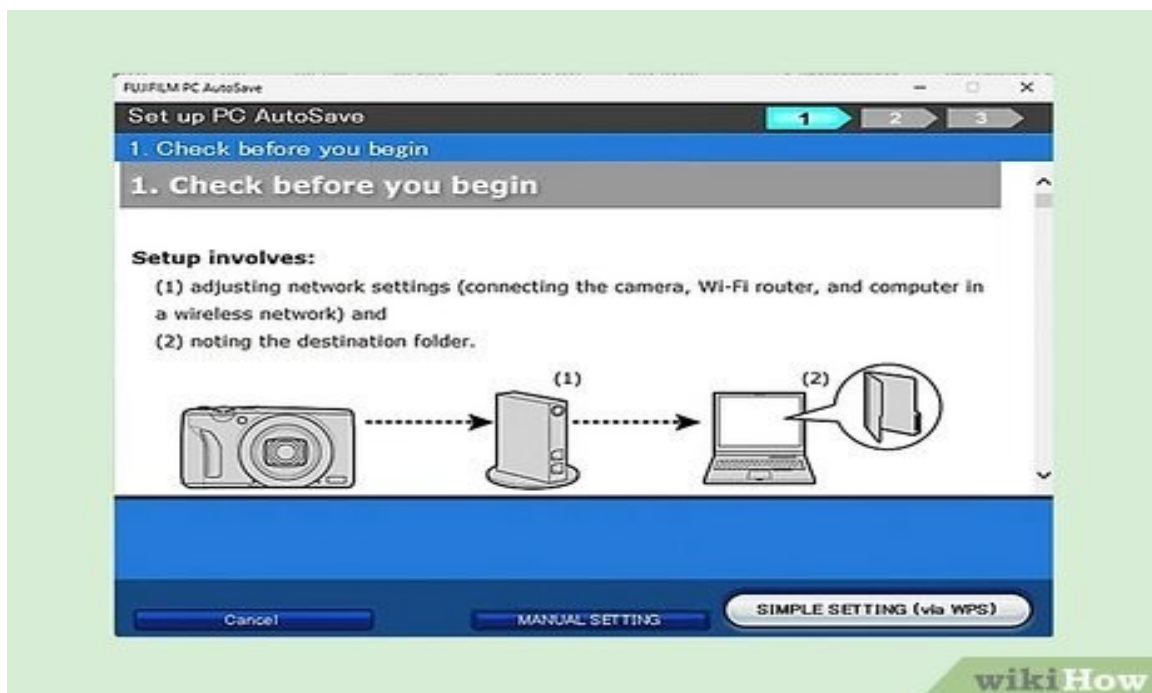
Requirement:

Computer system, Digital Camera, USB Wi-Fi adapter, USB Bluetooth adapter, USB Storage device, Projector, USB cables, Drivers.

A. Installation and Configuration of Digital Camera

Procedure:

1. Connect digital camera to computer using USB cable.
2. Switch on the camera.
3. System detects the device automatically.
4. Install driver if required.
5. Open **This PC** → Camera storage.
6. Access or transfer photos/videos.



Result:

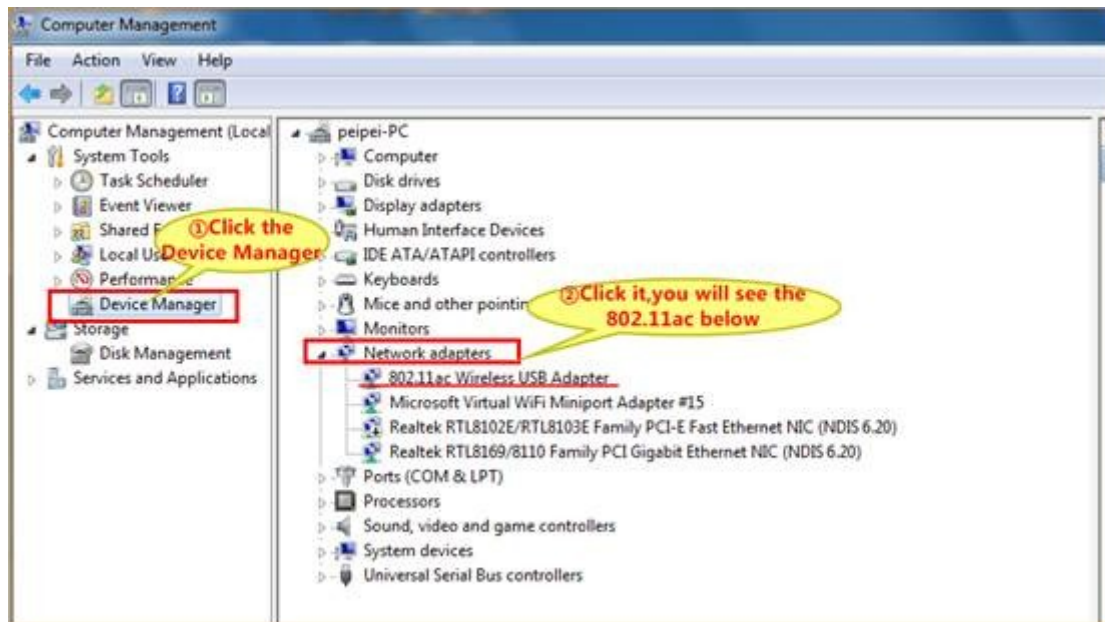
Digital camera is successfully connected and configured.

B. Installation and Configuration of USB Wi-Fi Adapter

Procedure:

1. Insert USB Wi-Fi adapter into USB port.
2. Install driver from CD or website.
3. Restart system if required.
4. Click on **Network icon**.

5. Select Wi-Fi network and enter password.
6. Click **Connect**.



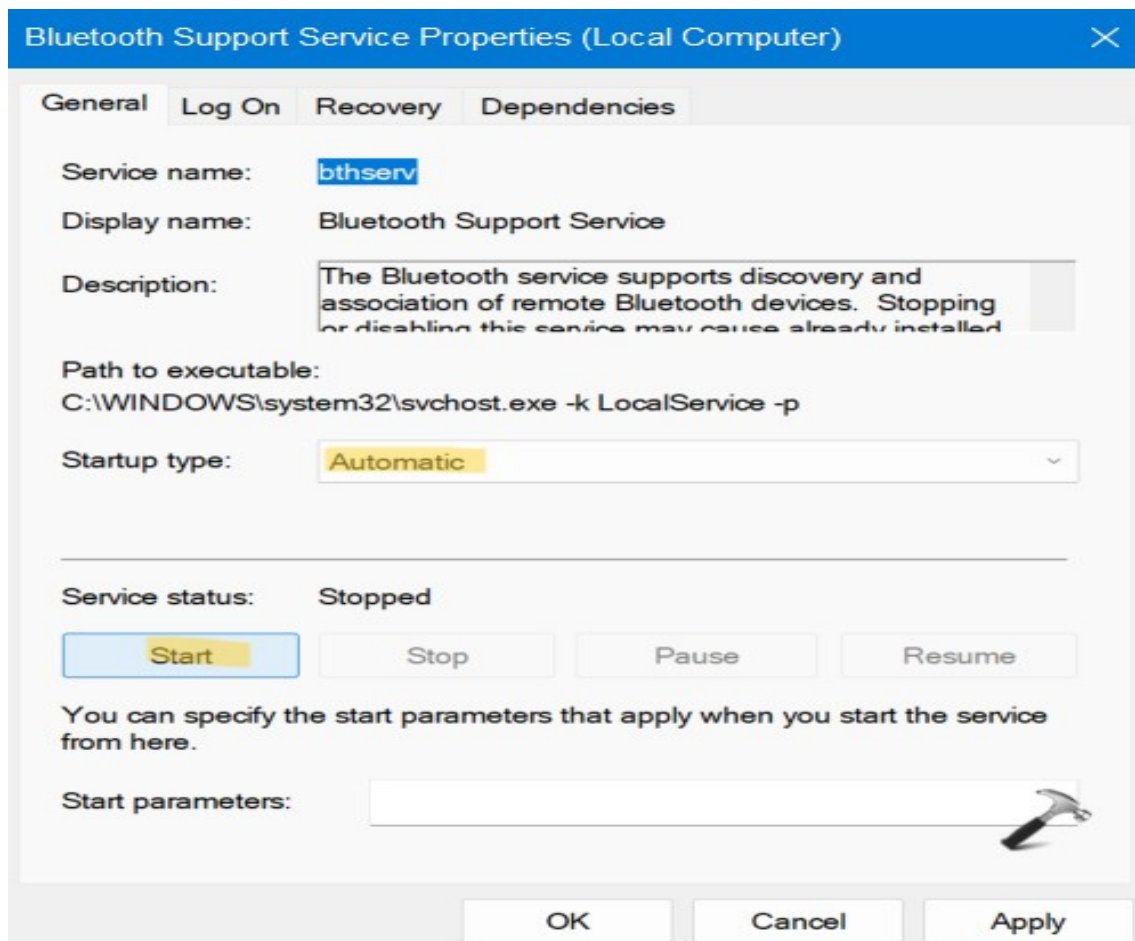
Result:

USB Wi-Fi adapter is installed and internet is connected.

C. Installation and Configuration of USB Bluetooth Adapter

Procedure:

1. Insert USB Bluetooth adapter into USB port.
2. Windows installs driver automatically.
3. Turn on Bluetooth from settings.
4. Click **Add Bluetooth Device**.
5. Pair with required device (mobile, headset, etc.).



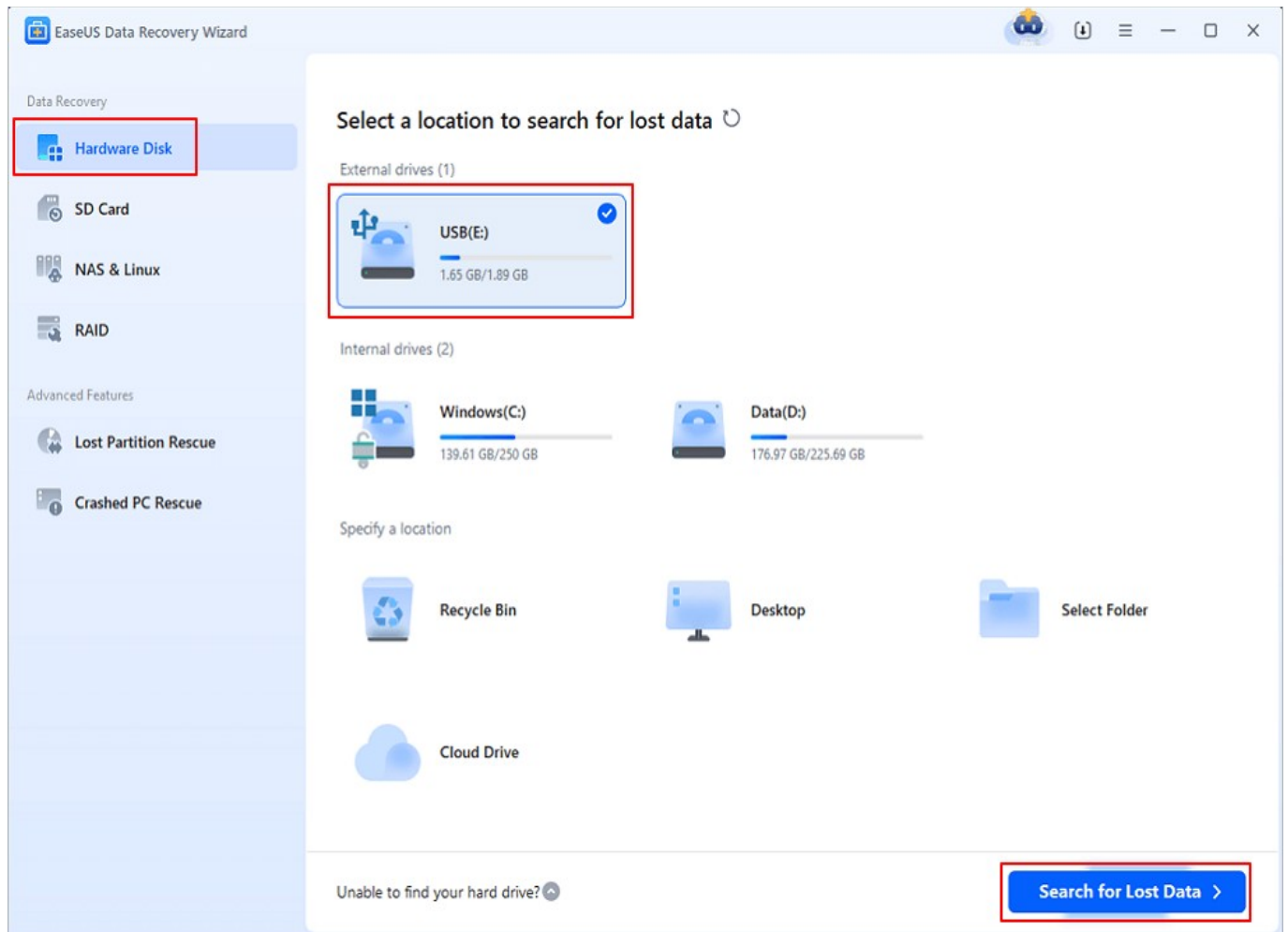
Result:

USB Bluetooth adapter is installed and working.

D. Installation and Configuration of USB Storage Device

Procedure:

1. Insert USB pen drive or hard disk into USB port.
2. System detects device automatically.
3. Open **This PC**.
4. Access, copy, or save data.
5. Safely remove device after use.



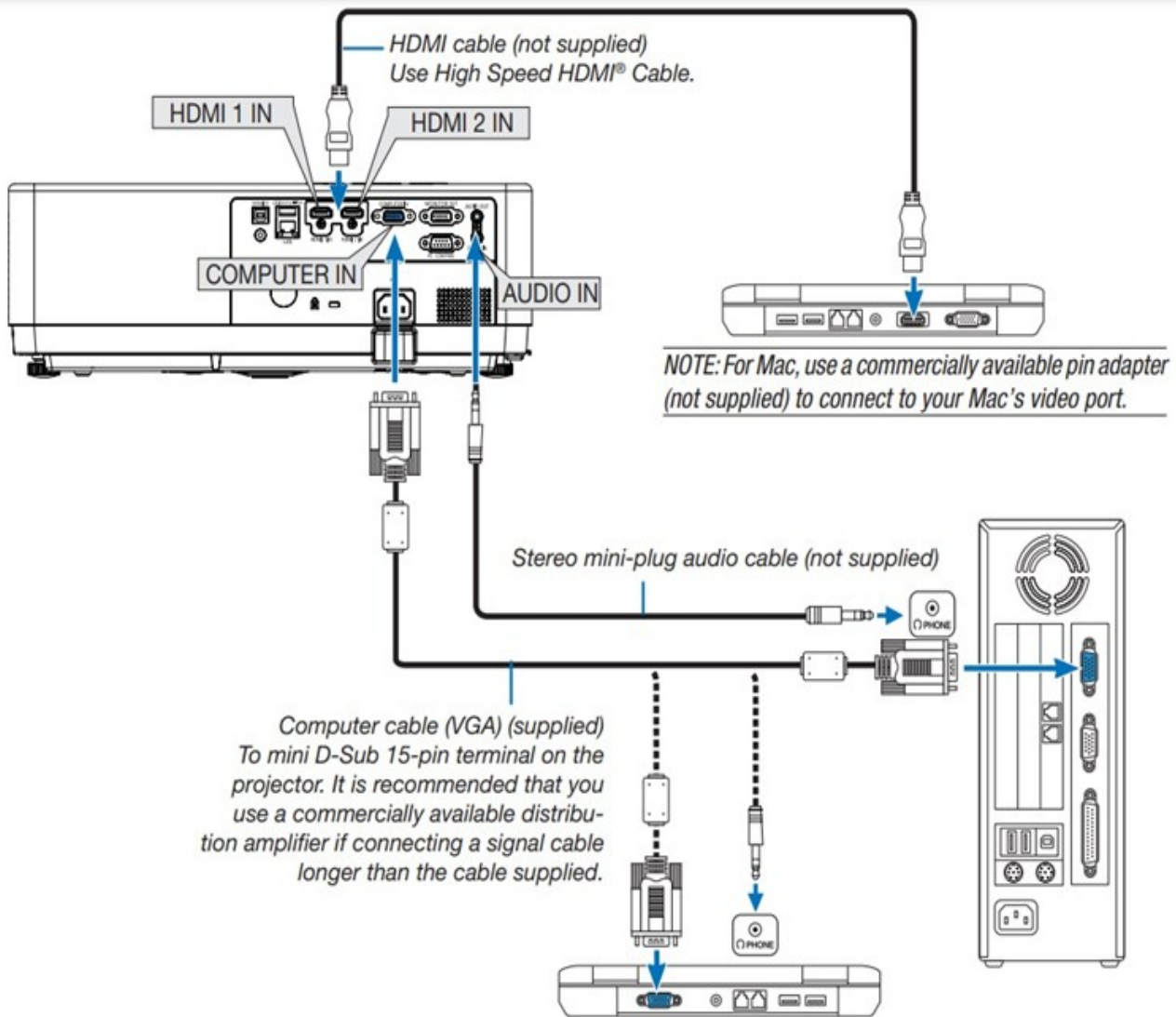
Result:

USB storage device is ready for data transfer.

E. Installation and Configuration of Projector

Procedure:

1. Connect projector to computer using HDMI/VGA cable.
2. Switch on projector and computer.
3. Select correct input source on projector.
4. Press **Windows + P**.
5. Choose display mode (Duplicate / Extend).
6. Adjust resolution if needed.



Result:

Projector is successfully connected and displaying output.

Conclusion:

Proper installation and configuration of I/O devices ensure smooth data transfer, connectivity, and display.

8. Multiple OS loading and trouble shooting

Aim:

To install and load multiple operating systems on a single computer and troubleshoot common boot problems.

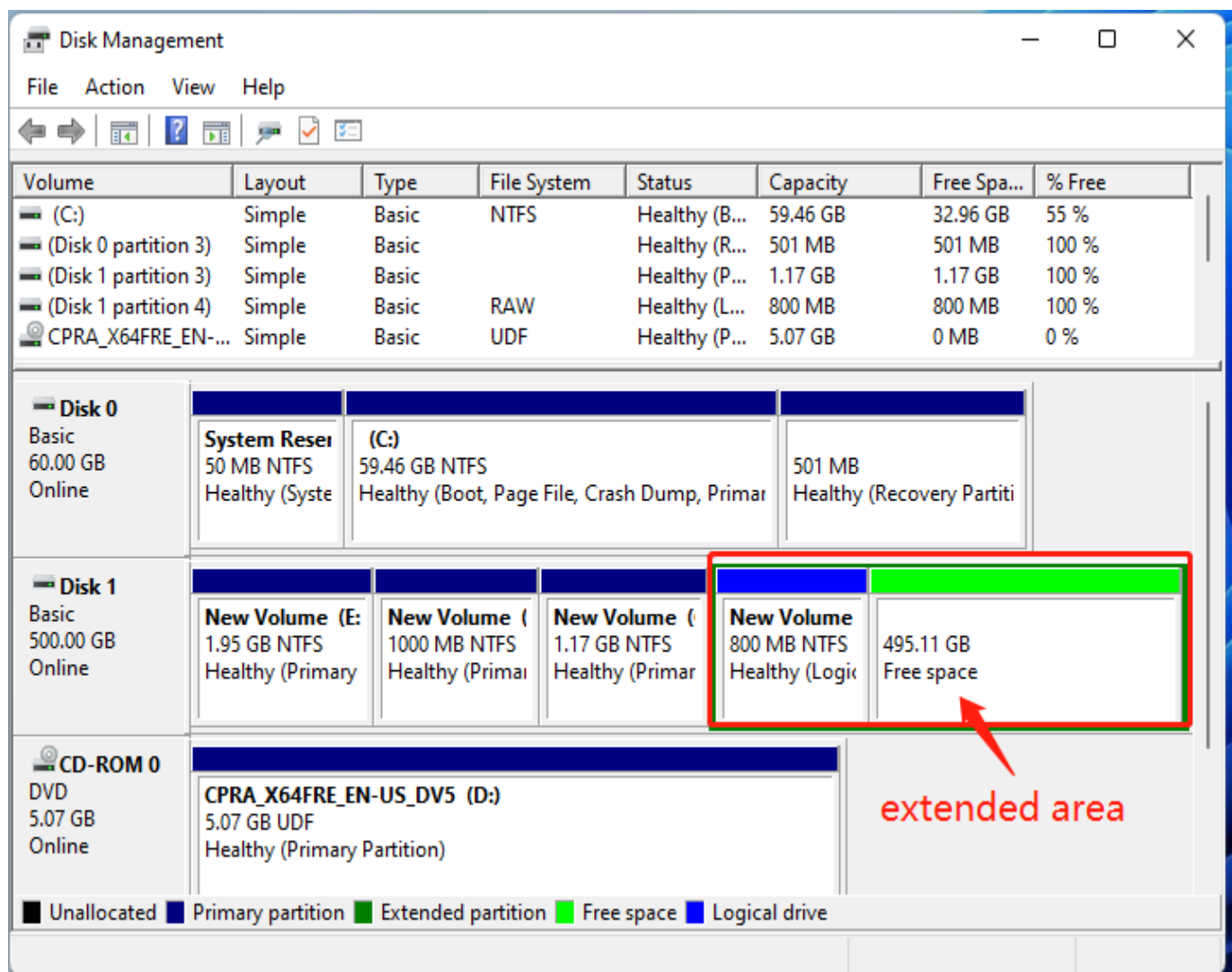
Requirement:

Computer system, Two or more OS (Windows / Linux), Bootable USB or DVD, Hard disk.

A. Multiple OS Loading (Dual Boot Installation)

Procedure:

1. Create free space on hard disk using **Disk Management**.
2. Insert bootable USB/DVD of first OS.
3. Install the first operating system (e.g., Windows).
4. Insert bootable USB/DVD of second OS (e.g., Linux).
5. Select **Install alongside existing OS** option.
6. Choose partition for second OS.
7. Complete installation and restart system.
8. On boot, a **Boot Menu** appears to select OS.



Result:

Multiple operating systems are successfully installed on one system.

B. Troubleshooting (Common Problems & Solutions)

Problem	Solution
OS not booting	Repair boot using recovery media
Boot menu not showing	Set correct OS in BIOS/Boot loader
Slow boot	Disable unnecessary startup programs
Driver not working	Install correct device drivers
System crash	Run system repair or safe mode
Wrong OS loading	Change boot priority in BIOS

Conclusion:

Multiple OS loading allows users to work on different operating systems on one computer. Troubleshooting helps in fixing boot and system errors.